

**Ahmedabad
University**

Stroop Task Experiment

PSY310: Lab in Psychology

Lab Report

Date: 10th September 2026

Name of the Student: Aayushi Dave

Enrollment No: AU2420101

GitHub Link: <https://github.com/aayushi111525/Stroop-Task-Experiment>

Introduction:

The Stroop task experiment is very much used in experimental psychology. Even after ninety years, it is still being used to do further research in areas like attention, cognition control etc. Stroop task usually measures how a written word which is an irrelevant information interferes or conflicts with a relevant information which is the ink color in which it is printed). This interference happens because when we read a familiar word, it becomes automatic processing whereas naming an ink will be more of a deliberate processing. The interference is tested by comparing Response Times (RT) between two types of trials which are congruent and incongruent trials. Congruent trials is where the words and the ink color match, whereas in the Incongruent trials, they don't match.

Method:

In the experiment, there were two types of trials, i.e. congruent and incongruent. The participant was asked to focus on the ink color and press 'r', if they saw red ink and 'b' if they saw blue ink. There were a total of 10 trials, with 4 conditions. The conditions were presented in random order and the accuracy and reaction time was recorded for every trial.

Participants:

There was 1 participant in this experiment who was a 20- year old male. The participant was totally healthy and had normal vision. The participation was voluntary and consent was sought before starting the experiment.

Materials and Stimuli:

The experiment was run on a Windows laptop using PsychoPy according to the professor's tutorial video. Stimuli consisted of two words, 'RED' and 'BLUE' which were presented in red or blue font color. The background was grey and there was a fixation (circle) on the centre of the screen.

Image 1:

This is a Incongruent Stroop Stimulus trial. You can see the word 'BLUE' which is displayed in red font color. This will cause interference because of the difference in the semantic meaning (word) and the feature (font color).

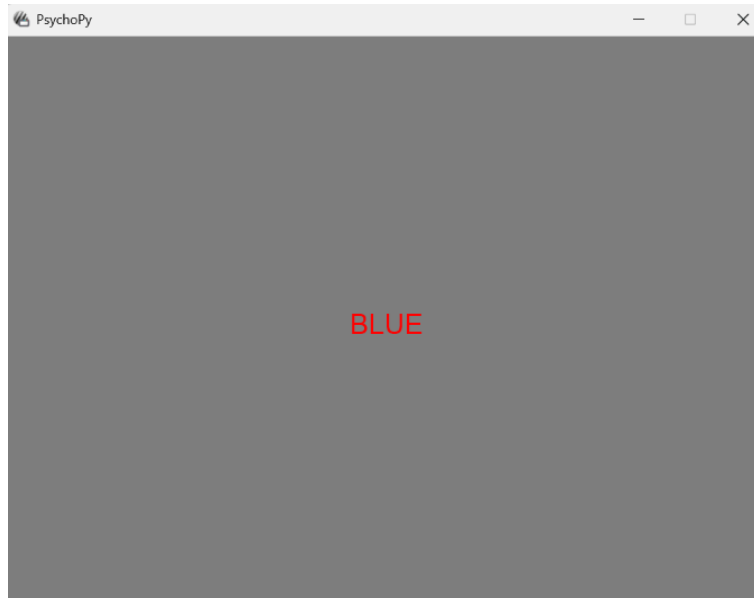
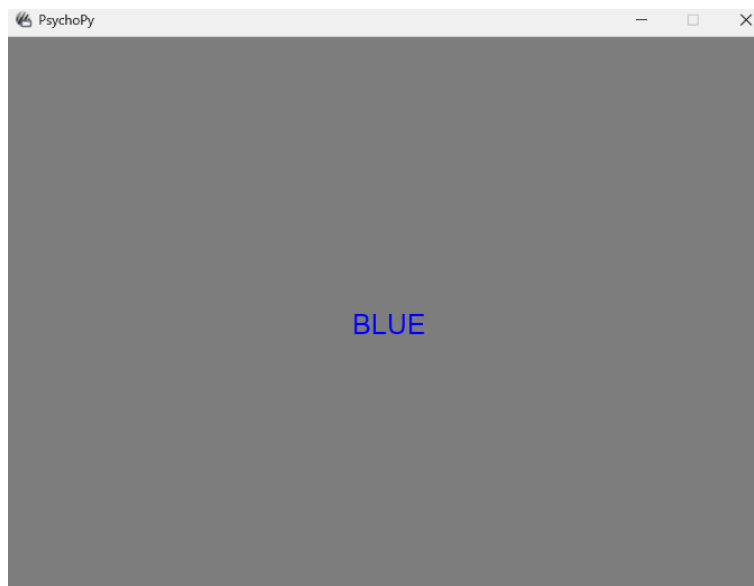
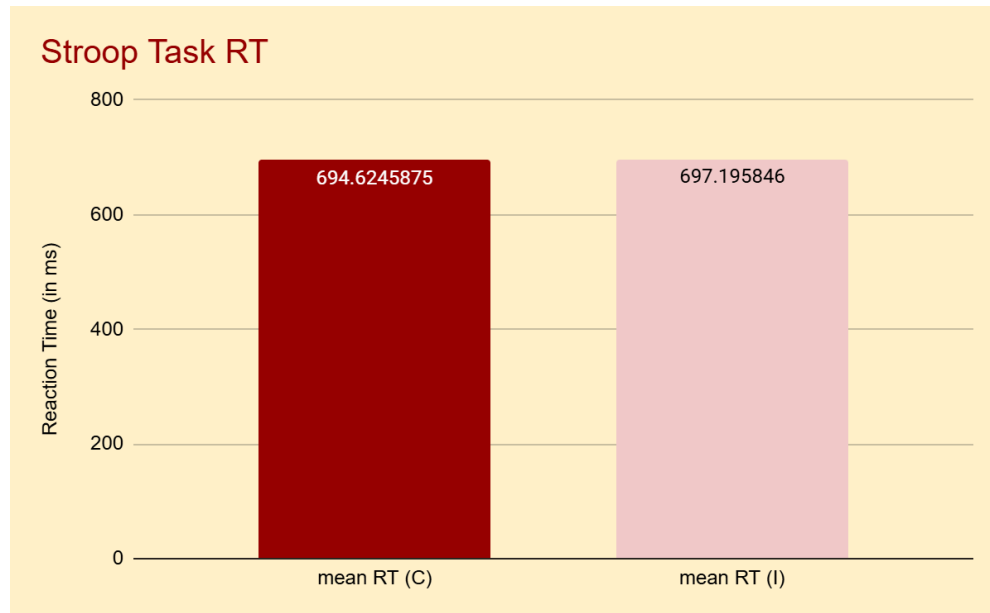


Image 2:

This is a Congruent Stroop Stimulus trial. You can see the word 'BLUE' which is displayed in blue font. In this case, the assumption is that the response will be faster because of the similarity in the semantic meaning and the color match also.



Results:



From this graph, we can see that the mean reaction time for congruent trials was 694.62 ms and 697.20 ms for the incongruent trials and the difference that we get from this is only about 2.57 ms. This gap is very small. The congruent and incongruent response times were nearly identical.

Discussion:

This result is very surprising because it sort of shows that there is a very negligent stroop effect. The stroop effect says that the incongruent trials would have more response time than the congruent trials because there is a conflict in the word meaning and the ink color. This case here goes against what generally the results of stroop effect should be. This could mean that the interference was very less for the participant who performed this task.

Limitations:

There was only one participant and because of this we cannot generalise the result because this cannot be said to have absence of interference. Only two colors and two words were used, so this simplified version of this stroop paradigm that might not capture the interference effect fully.

References:

Weir, Kirsten. "The Stroop Effect's Long (and Colorful) Influence." *Apa.Org*, 24 Mar. 2025,

<https://www.apa.org/research-practice/conduct-research/stroop-effect>

"Stroop Experiment." *Psych.Hanover.Edu*,

https://psych.hanover.edu/javatest/cle/Cognition_js/exp/stroop.html